

Insect Navigation: Brains as Symbol-Processing Organs

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1. Introduction

Digger wasps dig nest burrows in which they lay an egg. They cover the burrow entrance with pebbles, making it all but invisible, then fly off in search of insect prey. They sting their prey to paralyze it, carry it back to the burrow, open the burrow, drag the prey inside, exit the burrow and close it up again. The larva that hatches from the egg eats the paralyzed, still living prey. This aspect of the natural history of digger wasps cautions us against an overly sentimental view of the workings of mother nature, at least at the insect level. Quite aside from pointing up this moral, it raises two questions of profound interest to contemporary cognitive scientists and neurobiologists: (1) How does the wasp find her way back to her invisible nest? (2) What does the answer to that question tell us about the nature of the brain?

1.1 Behavior suggestive of a cognitive map

Tinbergen and Kruyt (1938) began the process of answering these questions when they arranged a circle of sticks and pine cones around the nest while a wasp was digging it, then displaced the circle while she was gone. The returning wasp landed in the center of the displaced circle. She could not find her nest, even though it was only a meter away. Their experiment demonstrated that the wasp located the entrance to it by reference to nearby landmarks, not by the sight or smell, etc. of the nest itself. This is but one example of the quite general principle that animals locate their goals by reference to the surrounding terrain, and not, contrary to what people commonly assume, by homing on a sensory beacon emanating from the goal itself (see Gallistel, 1990, Chap. 5, for examples in animals ranging from bees to humans).

The discovery that the wasp locates her nest by reference to the landmarks or terrain features that surround it implies that she

notes the position of her nest relative to these landmarks before her departure, and, in fact, it has since been shown that the departing wasp makes a special survey flight for that purpose (Zeil, 1993a) . Moreover, it implies that when she returns she uses the remembered position of her nest relative to those landmarks to guide her flight path to a landing site. A record of the position of one or more points of interest relative to the positions of landmarks or salient terrain features is a **map*. A map that resides in the brain of an animal rather than on a sheet of paper or in the memory banks of a computer is called a **cognitive map*. Thus, the results of Tinbergen and Kruyt's experiment suggest that the wasp makes a cognitive map of the immediate vicinity of her nest and uses that map to choose a landing site.

The Tinbergen and Kruyt (1938) experiments did not, however, shed light on how the wasp navigated over the large distances covered by her foraging. They did not show how she found her way back from the place where she captured her prey to the vicinity of her nest, where the small-scale terrain features surrounding it would first become clearly visible. In 1950, Thorpe did some experiments that suggested that digger wasps relied on a cognitive map of their large-scale environment to find their way to the vicinity of their nest from wherever they happened to capture prey. He worked with a species of digger wasp that preys on caterpillars too large to be carried home in flight. The female of this species drags her prey home across the gravelly ground. In the experiment shown in , Thorpe found a wasp dragging her prey homeward. He first placed obstacles in her path that forced her to deviate from her course (numbered marks on figure). As soon as she had cleared each obstacle, she resumed her course. This proved that the wasp was not marching blindly along like a wind-up toy tank, without regard to where her marching was actually getting her. When forced to deviate from her course, she corrected for the deviation she had made --a result that has been confirmed recently by similar experiments with ants (Schmidt, Collett, Dillier & Wehner, 1992).

Next, Thorpe picked up the wasp, placed her in a box, and carried her about 30 meters to the east northeast of the capture point (broken line on Figure 1), where he released her. At the release point, a clump of grass obscured the view of her nest site. It is, in any event, unlikely that the view of the nest site plays any role in controlling most of the wasp's homeward journey because she

views the world from a height of a few millimeters. This means that her view of level ground more than a fraction of meter distant is generally obscured by small bumps in the terrain. Although at the release site she could not see or otherwise detect the area immediately around her nest, the newly released wasp nonetheless headed straight through the clump of grass on a course that lay within 15° of the true course from the release site to her nest. In other words, she set her course for a site whose position she could not observe. She did so in all likelihood by reference to those parts of the terrain that she could observe—such as the small birch trees (tree symbols on Figure 1) and the horizon contour formed by the rim of the depression (indicated by the hatching on the periphery of the map of the experimental terrain in Figure 1).

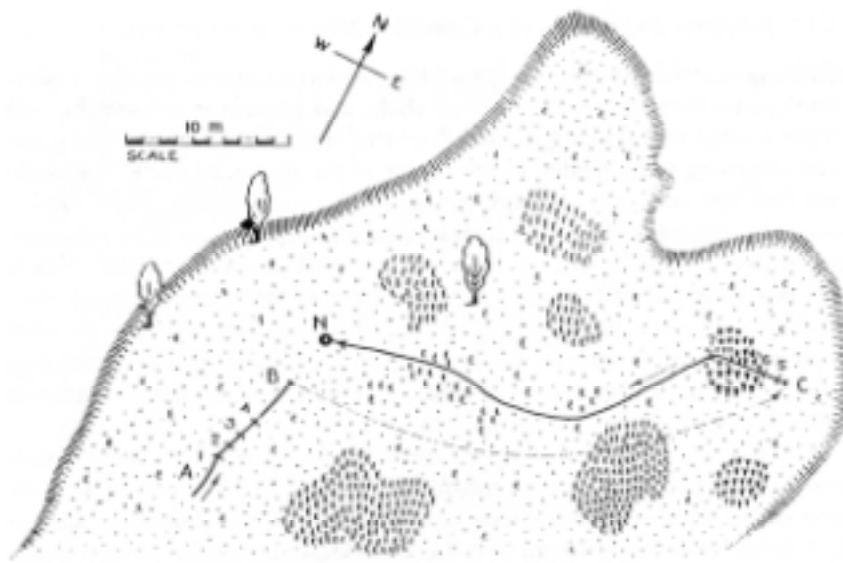


Figure 1. *Digger-wasp capture and release experiment. Heavy line indicates the course of the insect. It was first observed at A. Numbers 1-4 indicated where obstacles were placed in its path. It went round the obstacles and resumed its course. At B it was captured and carried to C (broken line). Numbers 5-7 indicate places where obstacles were again placed in its path. N indicates its nest. The hatching around the perimeter of the map indicates a slight depression in an area of gravelly heath land. Tree symbols indicate small birch trees. Patches of scattered plants of Erica and Calluna indicated by the 'E' symbol. Tufts symbolize tussocks of juncus grass. (From Thorpe, 1950, by permission of E.J. Brill, publisher.)*

1.2 Maps and navigational computation

The ability to use currently perceived terrain features to set a course for an area you cannot presently perceive generally requires the use of a map (also called a chart), which specifies the position of your destination relative to the positions of the landmarks you currently perceive. It also implies the ability to do navigational computations utilizing the data on the map, because navigation (course-setting) requires computation. Thus, these results suggest the brain of an insect has computational capabilities analogous to those found in, for example, a GPS (Geographic Position Sensor).

A GPS is a miniature marvel that has recently become available to the general public in a package that weighs a few ounces and fits in the palm of the hand. It consists of a computer specialized for navigational computations, a precision clock, and a radio receiver tuned to the signals broadcast by an array of earth-orbiting navigation-satellites. The satellites provide signals from which it is possible to compute the position of a receiver--the latitude, the longitude, and the elevation of the GPS--to within about 100 feet.

The brain of the GPS is its computer. You key into the long-term memory banks of the computer the latitudes and longitudes (and, if you like, also the elevations) of ten to several hundred points of interest to you--harbor entrances, buoys, headlands, the ends of runways, trailheads, turn-off points, campgrounds, etc--identified by a keyed-in name or number. These points of interest are called waypoints. Their latitudes, longitudes and elevations constitute a map. Thereafter, when you key in the name or number of a waypoint, the GPS uses the waypoint's latitude and longitude (retrieved from its memory banks) and the latitude and longitude of the GPS's current position (computed second by second from the radio signals it is receiving from the satellites) to compute the bearing and range of the waypoint. The **bearing* is the compass direction of the waypoint, for example, of the entrance of the harbor you are making for. The bearing tells you the course to steer (assuming that the waypoint in question is not more than a few hundred miles distant). The **range* is your distance from the waypoint. If you ask it, the GPS will also compute for you the range and bearing of any waypoint from any other, as well as things like your speed, and your estimated time of arrival.

1.3 Representational (symbol-processing) and non-representational (sub-symbolic) theories of mind

The computer in a GPS computes courses and distances to be traveled from its map, that is, from the positions of points of interest recorded in its memory banks. The fact that the digger wasp can set off on an appropriate course for a destination she cannot currently perceive after she has been displaced to a known point in her environment suggests that she, too, can compute courses from her cognitive map. What makes this of interest to cognitive scientists is that it suggests that the brain of an insect is a symbol processing organ. It suggests that the brain uses computation to construct a cognitive map and to compute courses from the positional values stored in that map. The positional data stored on a map together with the computations that utilize those data to set courses constitute a **representation*, a model of the world used to determine courses of action in that world. Thus, the homing abilities of insect might be taken as strong evidence in favor of a computational-representational theory of mind.

Computational-representational theories of mind are at the heart of a controversy that has raged through psychology, computer science, linguistics and neurobiology throughout much of this century. The community of cognitive, behavioral and neural scientists has long been divided over the issue whether the brain really represents the world within which the animal acts or merely appears to do so. Setting out toward destinations you cannot see suggests that you know where those destinations are relative to where you are, just as speaking a language fluently suggests that the speaker knows its grammar, but many participants in this debate believe that these conclusions are naive, that models positing symbolic knowledge inside the head are doomed to fail. They believe that behavior that suggests knowledge of the world is produced, not by a brain that computes symbolic representations of the world, but by a vast and plastic neural network that does not contain within it any symbolic knowledge of the world.

The debate between those who believe that much behavior is mediated by symbolic knowledge of the world in the brain and those who believe that it is not has been going on for a long time, but recently, this question has been formulated in the following terms: Is the brain a symbol-processing system? The rise of cognitive psychology in the sixties reflected the emergence of the (then novel)

view that the brain was a symbol-processing system analogous to a computer, and that it used its symbol-processing capacity to construct representations of the environment in which it functioned. This cognitive conceptual framework contrasts with the earlier framework of the behaviorist school, in which the brains of animals were not symbol-processing systems and did not construct representations of their environment.

The best known proponent of the behaviorist school of psychology were Clark Hull and B. F. Skinner. Although they were all contemporaries, psychologists like Hull and Skinner did not in those days pay much attention to the work of ethologists like Tinbergen and Thorpe, so I do not believe that either Skinner or Hull ever offered a non-representational account of performances like that shown in Figure 1. Hull (1930), in an immensely influential paper entitled "Knowledge and Purpose as Habit Mechanisms," offered an account of what he supposed to be the process by which a rat traversed a simple maze to reach food in a location where it had previously found food. This account was intended to illustrate Hull's contention that the rat's behavior did not depend on a representation of the position of the food within the spatial framework defined by the maze and the surrounding room and did not involve anything like the navigational computations that go on in a GPS. The essence of Hull's account was that the rat developed a chain of simple "habits" or stimulus-response units, which were in essence conditioned reflexes. Each individual habit formed because that particular response made to whatever stimuli impinged on the animal at that point in the maze, had repeatedly led to reinforcement, whereas other responses made to those stimuli had not. The reinforcement selectively strengthened the conducting pathway in the nervous system leading from the receptors excited by a particular stimulus configuration to the muscles whose contraction and relaxation generated the movement of the animal (the response). The animal's movements at different points in the maze were not controlled and integrated by a representation of a course to be followed. In fact, each response was a little independent atom of behavior. The only thing that integrated responses in different parts of the maze were their stimulus consequences. The stimuli generated by a preceding response were often an important part of the stimulus configuration for the next reflex. Also, responses the animal originally made only at sight or smell or taste of the food (e.g., salivation) could, Hull argued, come in time to be made early on in the maze (anticipatory goal

responses) and their stimulus consequences (the feel of salivation in the mouth) could become part of the stimulus configuration that elicited a movement in the maze, such as turning right at a choice point. The whole point of Hull's argument was that the rat did not have knowledge of the spatial layout of the maze and the location of the food (a cognitive map), it merely appeared to have such knowledge. And this was precisely the conclusion that Tolman (1948) questioned in his famous paper entitled, "Cognitive Maps in Rats and Men." (Tolman also paid no attention to the ethologists, even though their work provided more powerful evidence for his thesis than did his own experiments.)

The computational-representational or symbol-processing conception of the brain processes that generate behavior also contrasts with the conceptual framework urged by those proponents of neural net modeling who argue that the brain is not a symbol-processing organ, and that models of psychological function based on the assumption that it is have and must fail. Both the behaviorists and the champions of neural net models have argued that behaviors such as navigating through an environment or learning to speak a language make it appear that we have formed a representation of the spatial form of the environment (a cognitive map) or of the grammar of the language (a mental grammar), when in fact we have not. In this chapter, I will review experimental data on the navigational abilities of insects and argue that these data strongly support the view that brains are in fact symbol-processing organs.

1.4 Symbol-processing systems

A symbol-processing system, such as a conventional computer, has two fundamental components: First, it has a set of symbols, for example, symbols for distances, symbols for positions, symbols for temporal intervals, symbols for numerosity, symbols for verbs, and so on. The symbols take on different values at different times, depending on the values of the variables that they denote. For example, the symbol for the distance between oneself and home takes on different values depending on how far one is from home. In thinking about symbols, it is essential to distinguish between the symbol itself--a mark on paper, a bit pattern in a digital computer, or a voltage in an analogue computer, representing a distance or an interval--and the variable that it denotes--the actual distance or the actual temporal interval. The essence of a symbol is that it denotes or refers to something other than itself. Second, a symbol-

processing system has a set of operations that may be performed on the values of the symbols, for example, the operation of adding value of one symbol to the value of another to generate a third value. If the symbols being added represent temporal intervals, then the value that is their sum represents the combined interval or sum of the intervals. Thus, symbols may be combined to generate either symbol strings or new symbols, whose values are determined by the values of the symbols that were combined and by the combinatorial operation (e.g., addition or subtraction or multiplication or exponentiation, etc.).

The use of symbols in conventional computation is sometimes said to be **compositional*, because the computation may be composed out of any of the values that the symbols happen to have regardless of how they came to have those values. The addition process (or operation) simply combines two values to generate a third value. In Figure 2, Calvin shows a proper appreciation of the deep mystery of this operation, which is too often taken for granted. Often, the values used in a particular addition operation are generated by earlier operations, and this leads to the fundamental idea of the composition of operations or functions, which the student will have encountered both in calculus and in computer programming. The notion of composition of functions is expressed algebraically as follows: If $f(x) = x + a$ and $g(x) = x^2$, then $f(f(x)) = (x + a) + a$ and $g(f(x)) = (x + a)^2$ and $f(g(x)) = x^2 + a$, and so on. In other words, because the operations (in this case the functions f and g) are not affected by the origins of the values they operate on, you can take the values generated by one of the operations and make it the subject of the other operation or of a repetition of the first operation.

Calvin and Hobbes

by Bill Watterson

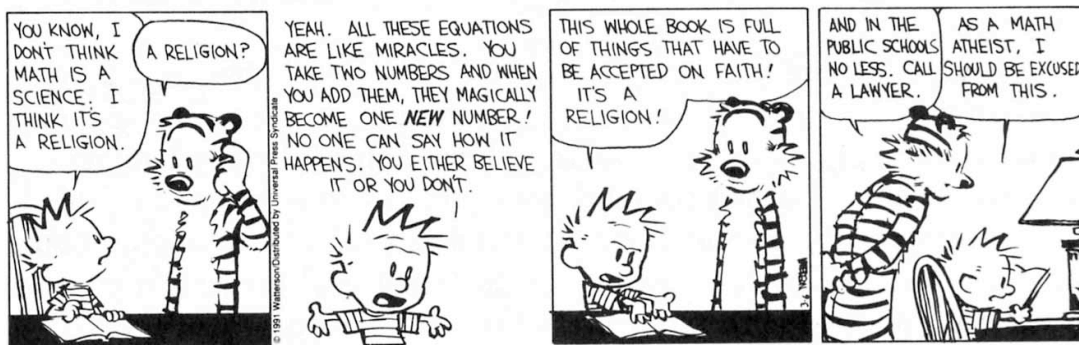


Figure 2 *Calvin appreciates the essential mystery of the fundamental arithmetic operations. Neurobiologists are math agnostics: they do not understand how the nervous system combines value-specifying signals to generate new value-specifying signals in a manner that mimics the fundamental operations of arithmetic--addition, subtraction, multiplication, division, and ordination--and some doubt that it in fact does so. (Copyright by Bill Watterson and reproduced with his permission.)*

The values of variables need not come from previous computations, however. Sometimes the values of symbols come from sensing and measuring devices sensitive to the value of the real-world variable the symbol denotes or refers to. Thus, the values for symbols of temporal intervals may come from timers, the values for time symbols from clocks, the values for distance symbols from a radar ranging device, the values for a direction symbol from a magnetic compass, and so on. A symbol-processing device requires a mechanism or a procedure that establishes which symbols denote which real world variables, and it generally requires calibration or checking procedures to insure that the values of those symbols do in fact accurately represent the values of the real-world variables to which they refer.

Supplying a value for a symbol is sometimes called **binding* a value to a variable. For example, in making a computation on a reverse-Polish calculator, you first key in one value and hit Enter, thereby binding the keyed-in value to the addend; then you key in another value (binding that value to the augend); and finally, you hit the '+' button to set the addition operation in motion. The electronic process that mediates the addition operation determines and displays a new value, which is the sum of the addend and the augend. (Calvin's problem is that he is not sure how much to credit what the addition operation does.)

The digital nature of contemporary calculators is irrelevant in the above illustration. The use of a slide rule to multiply serves as an equally good example¹. First, you move the initial mark (index

¹It is a painful reminder of my advancing age and the rapidity of technical progress that I must tell the contemporary student what a slide rule is or was. It was an analog calculator used by technical people for multiplying and dividing until quite recently, when it was superseded by the incomparably more powerful pocket calculator. It used logarithmic scales marked on both a frame and a slider. The slider, with its scale, moved within the frame, with its scale. It exploited the fact that $\log(x*y) = \log(x) + \log(y)$. A good slide rule was a lovely instrument, much treasured by its possessor.

point) of the slider so that it is opposite the value of the multiplicand on the frame (binding the value of the multiplicand.) Then you find the value of the multiplier on the slider (binding the value of the multiplier). Finally, you note the value on the frame opposite the value of the multiplier on the slider. This is equivalent to hitting the 'x' button; in doing this, you perform the multiplication operation.

In both examples of the compositional nature of conventional computation, a human operator supplies the values to be added or multiplied (operated on) and reads the results, but that is also irrelevant. When conventional computers are used to control processes, as they are in modern factories, automobiles, and appliances, the values for the symbols are supplied by sensors and timers, and the results of their computations are used to control actuators. Human participation is not necessary. If the brain is a computer, then it is a process-control computer; it does not require an **homunculus* --a little man in the head who operates the computer.

Finally, the use of numbers to be the values of the variables is also irrelevant. In electronic analog computers, voltages specify the values of variables such as wind velocity, roll-of-the ship, and so on. These voltages are fed into operational amplifiers configured either to add or to multiply their input voltages in producing their output voltage. What is crucial is that there is some machinery whereby the values can be bound to variables (symbols) within the symbol-processing device and whereby these values may then be combined with other values to generate still other values that may themselves enter into further computation. It is also crucial that there is some machinery that establishes what it is that a symbol denotes.

1.5 Neural nets

The compositional property is so obvious an attribute of conventional computation that the student may wonder why it is singled out for comment. I single it out because this property is missing in neural net "computation," or at least in many of the best known models of this style of computation (see Rumelhart & McClelland, 1986). The rise of neural net theories in the eighties was in part a reaction to the view that the brain performed computations in the conventional (symbol-processing) sense. Some of the originators of the neural net approach to psychological

Many neural net theorists seek to eliminate this conceptual chasm between neurobiology and cognitive psychology by denying the assumption that the brain has a symbol-processing capacity, which is equivalent to the assumption that the brain computes in the conventional sense of that term. Our current inability to explain basic symbol-processing operations neurobiologically is also a major reason why neurobiologists tend to favor the view that the brain is not a symbol-processing organ.

The other possibility is that those mechanisms in the nervous system that mediate its symbol-processing capacity remain to be discovered or identified. If this is the case, then our current understanding of the nervous system is seriously deficient. The brain is the organ of mind. If the essence of mind is the processing of symbols, then we have not begun to acquire a serious understanding of that organ until we can account for its symbol-processing capacity.

In short, the stakes in the debate about whether the brain is a symbol-processing organ are large. Either much of current psychological theorizing rests on fundamentally erroneous assumptions about the nature of mental processes. Or, our understanding of the behaviorally important properties of the nervous system does not extend to those processes that are at the heart of all higher function. Small wonder that the debate sometimes grows heated.

1.7 Mental representations

Symbol-processing theories of mind and representational theories of mind are intimately related. Representational theories of mind maintain that the brain constructs mental representations and uses these representations in the decision processes that generate action. The mental representations exist in the form of computed values for symbols. Thus, for example, the mental representation of an animal's position consists of the value of its position vector, which is computed by its dead-reckoning mechanism (see below).

The definition of a mental representation can be given in a few well chosen words, but, like most definitions of complex concepts, the definition then requires extensive explanation. A **mental representation* is a functioning isomorphism between a set of processes in the brain and a behaviorally important aspect of the world. This way of defining a representation is taken directly from

the mathematical definition of a representation. To establish a representation in mathematics is to establish an isomorphism (formal correspondence) between two systems of mathematical investigation (for example, between geometry and algebra) that permits one to use one system to establish truths about the other (as in analytic geometry, where algebraic methods are used to prove geometric theorems).

An **isomorphism* is a relation between two systems. Each system is composed of entities and relations or processes involving those entities. Take for purposes of illustration a navigational computer as one system and the landmarks and goals represented on the computer's map as the other system. The position of the Los Angeles airport is represented in the computer by a number pair--the latitude and longitude values that, by convention, are used to represent positions on the surface of the earth in navigational computations. These number pairs are called **position vectors*. They are physically realized as bit patterns in the computer's RAM or by patterns of magnetization on its hard disk. The bit patterns in a RAM chip are not themselves positions in the world, they are bit patterns in a RAM chip. These vectors, or, if you like, the bit patterns that physically realize them, are the entities in one system, the **representing* system. The entities in the representing system are called *symbols*. The set of these entities, that is, the set of position vectors in memory constitute the computer's map. (To be useful, there must also be a list or table associating other information with these position values, e.g., the names of cities, but it is the position values that enter into navigational computations.) The corresponding entities in the other system--the **represented* system--are the actual positions themselves. These positions are, of course, not bit patterns, they are places in the world.

A navigational computer performs operations on position vectors. These operations yield answers to navigational questions, such as, Does Lake Superior lie on the great circle route between Los Angeles and Nairobi? These operations constitute the other essential aspect of a representing system--the operational or relational aspect. The corresponding aspect of the represented system is the set of geometric relations between places in the world. The three places just mentioned instantiate a collinearity relation because Lake Superior does lie on the great circle defined by Los Angeles and Nairobi.

The computer with its map (its set of position symbols) is the representing system. The actual places out there in the world, the places whose positions are symbolized inside the computer, constitute the other system. The two systems are isomorphic systems because the surveying procedures for determining numerical latitudes and longitudes and the program for performing navigational computations are together such that the computer says that three places lie on a great circle if and only if they really do. In this case, the computer says that Lake Superior lies on the great circle defined by Los Angeles and Nairobi if and only if the route that minimizes the distance flown on a trip from Los Angeles to Nairobi takes the airplane over Lake Superior. Finding the distance-minimizing route is the operation in the real world that is isomorphic to computing the great circle route. It is not easy to find the distance-minimizing route between Los Angeles and Nairobi by trial and error or any other readily performed non-computational real-world procedure, which is what makes computation so valuable in navigation.

The statement that Lake Superior lies on the great circle route if and only if the computer says it does not elevate the computer to the status of a reality-determining god. The if-and-only-if relation between the outcome of processes in the computer and the true state of the world arises only because of the care with which the surveying procedures and the navigation program are chosen and implemented. The if-and-only-if relation between the two systems is therefore fragile. It can fail with tragic consequences, as when the Korean Airlines 747 was shot down for flying over militarily sensitive portions of the Soviet Union. The plane, which was flying under the control of its navigational computer, was hundreds of miles off course as a consequence of a mistake in the after-take-off initializing input to the computer.

The navigational computations of animals also often fail, with fatal consequences. Thus, the processes in the nervous system that implement navigation are subject to intense natural selection. They are subject to natural selection because, like the computations of a navigational computer, they are **functioning* isomorphisms. The isomorphism between the representing system and the represented system is used by the representing system in coping with--surviving and reproducing in--the represented system.

1.8 Summary

A cognitive map is a mental encoding (symbolic representation) of the relative spatial positions of goals and landmarks. Its place symbols (position vectors in memory) are used in navigational computations, the processes that mediate the setting and holding of courses. If there is a cognitive map, then somewhere in the nervous system there must be neurophysiological states or activities that physically realize the position vectors that constitute the map (the neurobiological equivalent of the bit patterns in the RAM chip). These neurobiological position symbols are used in the neurophysiological processes that mediate the setting and holding of courses. If the cognitive map hypothesis is correct, then these neurophysiological processes will prove to be isomorphic to the computations that mediate conventional navigation in a navigational computer. That is, where the computation requires the addition of two variables (as does the dead reckoning computation to be described later), there will someday be found in the nervous system a process equivalent to addition. This process will take a neural signal that specifies the animal's cumulative displacement as of a moment ago and add to that a signal that specifies its most recent displacement to generate a signal that specifies its cumulative displacement as of right now. Thus, the use of a cognitive map, if it can be convincingly demonstrated, provides a paradigmatic example of the brain as a symbol-processing organ.

The advantage of studying animal navigation rather than other behavioral phenomena that have inspired intense debate between proponents of symbolic and non-symbolic analyses of mental function is that navigation is better understood from a computational standpoint than are most of the other areas in which this debate rages. Surveying and navigation have been intensively studied for thousands of years. While a proper humility in the face of possibly undreamed of limits to our understanding is always appropriate, it is nonetheless reasonable to believe that we may understand all there is to understand about navigation from a formal/computational standpoint. By contrast, our understanding of pattern recognition algorithms, language parsing algorithms, and language learning algorithms is rudimentary and incomplete. If we want to test whether the brain does or does not solve problems symbolically, it is best to work in an area where we believe that we may understand the "solution space," that is, the set of conceivable solutions.

2. Breaking the question down

The question whether insects have cognitive maps is too big a question to be addressed all at once. Like all major scientific questions, it needs to be broken down into a series of smaller, more readily answerable questions. Our extensive and long-standing understanding of geometry and navigation makes it possible to frame a series of specific questions.

The first two questions are can insects determine and remember angles and distances? Angles and distances are fundamental to the representation of position. If a device cannot determine angles and/or distances, then it cannot represent position; hence, it cannot have a map. On the other hand, if it can both determine and remember either angles or distances, then it has the basic ability to determine the values that define positions on a map. We can ask separately about angles and distances. However, it is important to know that from a formal standpoint, the two are intimately connected. Together, they constitute the metric properties of a geometry. (See Gallistel, 1990, Chap. 6 for a discussion of the grouping of geometric properties.)

The third question is whether insects do dead reckoning. Dead reckoning is the computation of net displacement (change in position) by either integrating velocity with respect to time, or by the discrete equivalent of this computation, which is adding up a series of small displacements to determine their sum. Dead reckoning is a fundamental mode of navigation because it enables one to move into unfamiliar terrain and still know the direction and distance back to home. By its very nature, it also generates a vantage-point-independent coding of positions. Thus, it may be used to relate one vantage point to another, which is crucial in computing a map of the terrain.

The fourth question is how insects hold a course? To hold a course, you must be able to continuously estimate which way is north. ("North" in this context just means some arbitrarily chosen but thereafter fixed reference direction.) One way of judging which way is north is by reference to a map of the terrain around you, which will identify for you the landmarks that lie to the north of your current position. Another way is by reference to the sun. Using the sun requires learning of the solar ephemeris function, which is the direction of the sun as a function of the time of day. Either of these methods implies the computation of a map of some kind.

The fifth question is how insects recognize a terrain from different angles and distances of view? The importance of this question is that in conventional navigation, such recognition is mediated by a map of the terrain. That is, knowledge of the three-dimensional lay of the land is used to interpret the two-dimensional projection of the terrain onto the retina. It is an open question whether it is possible to recognize the retinal projections of familiar terrains purely on the basis of the two-dimensional image, without computing aspects of the three-dimensional terrain of which the image is a projection. Thus, this question is intimately related to the question of whether insects have a map and, if so, what use they make of it.

The final question is whether insects have a integrated map, a map that enables them to compute the course between any two positions represented in their nervous system, whether or not they have ever traveled between those two positions in the past. If they do, then they have a map in the fullest sense of the term.

2.1 Can insects determine and remember angles?

Position on a plane may be defined by two distances (Cartesian coordinates), by an angle and a distance (polar coordinates), or by two angles (as in a three-point fix, see Figure 4). Thus, angles and distances are fundamental to the definition of position. If the nervous system cannot encode angles and distances, then it cannot encode position. If it can encode either angles or distances or both, then it can, in principle, encode positions. Thus, one fundamental question is whether the nervous system of an insect can encode an angle.

2.1.1. The dance of the honey bee symbolically expresses an allocentric angle. The discovery by von Frisch and his students of the dance language of the bees (von Frisch, 1967) provided exceptionally direct evidence that the nervous system of a bee could encode an allocentric angle, store the value of that angle in its nervous system, and use the stored value to control later behavior in which it communicated the value of the stored angle to other bees. When a foraging bee discovers a rich source of nectar and pollen, it records the angle formed by the position of the hive, the sun's azimuth, and the position of the food source. The sun's *azimuth* is the position on the horizon directly beneath the sun. An *allocentric* (other or elsewhere centered) angle is an angle that cannot be determined from sensory information available to an